

THE ALON–JAEGER–TARSI CONJECTURE

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Abstract

The Alon–Jaeger–Tarsi conjecture asserts that, for every finite field $\mathbb{F}_q$ with $q > 3$ and every matrix $A \in \text{GL}_n(\mathbb{F}_q)$, there exists a vector $x \in (\mathbb{F}_q^\times)^n$ such that $Ax \in (\mathbb{F}_q^\times)^n$. We survey the counterexamples over $\mathbb{F}_2$ and $\mathbb{F}_3$, the Alon–Tarsi proof for proper prime powers, and the Nagy–Pach theorem for sufficiently large prime fields. We also discuss connections with the permanent, permanent rank, and matrices similar to AJT matrices, and conclude with a rectangular-matrix analogue under a natural nonzero-row hypothesis.

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1 Introduction

Conjecture 1 (Alon–Jaeger–Tarsi). *For any prime power $q > 3$ and any $n \times n$ matrix A with $\det(A) \neq 0$ over the finite field $\mathbb{F}_q$, there exists a vector $x \in \mathbb{F}_q^n$ such that both x and Ax are nowhere-zero.*

This conjecture was proposed by N. Alon and M. Tarsi [3], and the case $q = 5$ was stated as an open problem by F. Jaeger [4].

Remark. If we replace $\mathbb{F}_q$ with an infinite field $\mathbb{K}$, the assertion is immediate. Since $\mathbb{K}^n$ cannot be covered by a union of finitely many proper subspaces, we can find a vector x such that both x and Ax are nowhere-zero.

2 Counterexamples over $\mathbb{F}_2$ and $\mathbb{F}_3$

This conjecture is known to be false for $q = 2$ and $q = 3$. In this section, we give the counterexamples.

For $q = 2$, if $n \geq 2$, we consider the matrix A over $\mathbb{F}_2$ as follows

$$A = (a_{ij})_{1 \leq i, j \leq n} := \begin{pmatrix} 1 & 1 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix},$$

where all the entries are zero except for the main diagonal and a_{12} . A direct calculation gives that $\det(A) = 1$ in $\mathbb{F}_2$. Since any nowhere-zero vector $x \in \mathbb{F}_2^n$ must be the all-ones vector, the first component of Ax is exactly zero.

For $q = 3$, if $n \geq 2$, we consider the matrix A over $\mathbb{F}_3$ as follows

$$A = (a_{ij})_{1 \leq i, j \leq n} := \begin{pmatrix} 1 & 1 & \cdots & 0 \\ 1 & 2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix},$$

where $a_{11} = a_{12} = a_{21} = a_{ii} = 1$, $3 \leq i \leq n$ and $a_{22} = 2$. A direct calculation gives that $\det(A) = 1$ in $\mathbb{F}_3$. If Ax is nowhere-zero, then

$$x_1 + x_2 \neq 0, \quad x_1 + 2x_2 \neq 0.$$

Examining the four possible pairs shows that there is no vector $x \in \mathbb{F}_3^n$ such that both x and Ax are nowhere-zero.

3 Known Results

3.1 Proper Prime-Power Fields

We record the polynomial argument of Alon and Tarsi [3]. Let $A = (a_{ij}) \in M_n(\mathbb{F}_q)$ be nonsingular, where $q = p^k$ and $k \geq 2$. Consider the polynomial

$$P_A(X_1, X_2, \dots, X_n) := \prod_{i=1}^n X_i \prod_{i=1}^n \left(\sum_{j=1}^n a_{ij} X_j \right) \in \mathbb{F}_q[X_1, X_2, \dots, X_n].$$

For $x \in \mathbb{F}_q^n$, we have

$$P_A(x) = \prod_{i=1}^n x_i (Ax)_i,$$

so $P_A(x) \neq 0$ is precisely the desired conclusion.

Theorem 3.1. *The function defined by P_A on $\mathbb{F}_q^n$ is not identically zero.*

Proof. Define

$$Q_A(X_1, \dots, X_n) = \prod_{i=1}^n \left(\sum_{j=1}^n a_{ij} X_j \right) \in \mathbb{F}_q[X_1, \dots, X_n],$$

and

$$L = \left\{ \alpha = (\alpha_1, \dots, \alpha_n) \mid \alpha_j \in \mathbb{Z}_{\geq 0}, \sum_{j=1}^n \alpha_j = n \right\}.$$

For $\alpha \in L$, let A_α be the $n \times n$ matrix obtained from A by repeating the j -th column α_j times (with the columns taken in any fixed order). Let c_α be the coefficient of $\prod_{j=1}^n X_j^{\alpha_j}$ in the expansion of Q_A .

Claim 1: The coefficient c_α is related to the permanent of A_α by

$$\text{Per}(A_\alpha) = c_\alpha \prod_{j=1}^n (\alpha_j!),$$

where the factorials are considered modulo p since $\mathbb{F}_p \subset \mathbb{F}_q$.

Proof of Claim 1: Group the terms in $\text{Per}(A_\alpha)$ according to the induced map from the row set to the original columns of A . Every term contributing to c_α occurs exactly

$$\prod_{j=1}^n \alpha_j!$$

times, because the α_j copies of the j -th column may be permuted arbitrarily. ■

Claim 2: There exists at least one $\alpha \in L$ with all $\alpha_j < p$ and $\text{Per}(A_\alpha) \neq 0$.

Proof of Claim 2: Write

$$A = (v_1 \mid v_2 \mid \dots \mid v_n)$$

and let

$$B = A^{-1} = (b_{j\ell})_{1 \leq j, \ell \leq n}.$$

Since $AB = I$, the ℓ -th column e_ℓ of the identity matrix satisfies

$$e_\ell = \sum_{j=1}^n b_{j\ell} v_j, \quad 1 \leq \ell \leq n.$$

By the multilinearity of the permanent in its columns,

$$\begin{aligned} 1 &= \text{Per}(I) = \text{Per}(e_1, \dots, e_n) \\ &= \sum_{j_1, \dots, j_n=1}^n \left(\prod_{\ell=1}^n b_{j_\ell, \ell} \right) \text{Per}(v_{j_1}, \dots, v_{j_n}). \end{aligned}$$

For a tuple $(j_1, \dots, j_n)$, let

$$\alpha_r = |\{1 \leq \ell \leq n \mid j_\ell = r\}|, \quad 1 \leq r \leq n.$$

Then $\alpha = (\alpha_1, \dots, \alpha_n) \in L$. Since the permanent is invariant under permutations of columns,

$$\text{Per}(v_{j_1}, \dots, v_{j_n}) = \text{Per}(A_\alpha).$$

Grouping the terms according to their multiplicity vector α , we obtain

$$1 = \sum_{\alpha \in L} s_\alpha \text{Per}(A_\alpha)$$

for suitable coefficients $s_\alpha \in \mathbb{F}_q$.

If $\alpha_j \geq p$ for some j , then $\alpha_j! = 0$ in characteristic p . By Claim 1,

$$\text{Per}(A_\alpha) = c_\alpha \prod_{r=1}^n \alpha_r! = 0.$$

Hence

$$1 = \sum_{\substack{\alpha \in L \\ \alpha_j < p \text{ for all } j}} s_\alpha \text{Per}(A_\alpha).$$

Therefore at least one $\alpha \in L$ with $\alpha_j < p$ for all j satisfies $\text{Per}(A_\alpha) \neq 0$. ■

By Claim 2, $c_\alpha \neq 0$ for that α , because $\prod(\alpha_j!)$ is nonzero modulo p (all $\alpha_j < p$). Hence we have found a monomial in Q_A of the form $c_\alpha X^\alpha$ with nonzero coefficient and $\alpha_j \leq p-1$. Let $T = (\mathbb{F}_q^\times)^n$. Since

$$P_A(x) = \left(\prod_{i=1}^n x_i \right) Q_A(x)$$

for every $x \in T$, it suffices to prove that Q_A does not vanish identically on T . Reduce Q_A modulo $X_i^{q-1} - 1$ for $1 \leq i \leq n$. The distinguished monomial $c_\alpha X^\alpha$ is unchanged because

$$0 \leq \alpha_i \leq p-1 < q-1.$$

If another monomial X^γ reduces to the same monomial, then

$$\gamma_i = \alpha_i + t_i(q-1)$$

for some integers $t_i \geq 0$. Since Q_A is homogeneous of degree n ,

$$\sum_i \gamma_i = \sum_i \alpha_i = n,$$

and hence $\sum_i t_i = 0$. Thus $\gamma = \alpha$, so no cancellation occurs. Therefore the reduced polynomial is nonzero, and Q_A is nonzero at some point of T . □

Remark. This proof cannot be applied to the case $q = p$ because the reduction modulo $X_j^{q-1} - 1$ can cause cancellation of monomials. We use the fact $q-1 > p$ for $k \geq 2$ in the last step of the proof.

By Theorem 3.1, choose x with $P_A(x) \neq 0$. The displayed product defining $P_A(x)$ then has no zero factor, which means $x_i \neq 0$ and $(Ax)_i \neq 0$ for all i . Hence the Alon–Jaeger–Tarsi conjecture is true for $q = p^k$ with $k \geq 2$.

3.2 Sufficiently Large Prime Fields

Now we follow the work of Nagy and Pach in [5], which is formulated in terms of special subsets of $\mathbb{F}_p$. We first introduce the definitions of S_k -type and N_k -type sets, which are used to construct the additive combinatorial argument.

Definition 3.2 (S_k -type). Let $k \geq 1$ and $p \geq 2k+1$. A set $K \subseteq \mathbb{F}_p$ is called S_k -type if for every $a \in K$ there is a $d \in \mathbb{F}_p^\times$ such that

$$a + td \in K \quad \text{for every integer } -k \leq t \leq k.$$

Thus every element of K is the middle term of a $(2k+1)$ -term arithmetic progression contained in K .

Definition 3.3 (N_k -type). Let $k \geq 1$ and $p \geq 2k+1$. A set $K \subseteq \mathbb{F}_p$ is called N_k -type if:

- K is S_k -type;
- for every $b \in \mathbb{F}_p \setminus K$, there is a $d \in \mathbb{F}_p^\times$ such that

$$b + td \in K \quad (1 \leq t \leq k).$$

Equivalently, after adjoining b , the set $K \cup \{b\}$ contains a $(k+1)$ -term arithmetic progression whose first term is b .

Definition 3.4 (Extremal functions $S_k(p)$ and $N_k(p)$). $S_k(p)$ is the minimum size of an S_k -type subset of $\mathbb{F}_p$, and $N_k(p)$ is the minimum size of an N_k -type subset.

Remark. It is clear that $2k+1 \leq S_k(p) \leq N_k(p)$.

Definition 3.5 (Group ring). Let G be a finite group and R a commutative ring. The group ring $R[G]$ is the set of formal sums $\sum_{g \in G} r_g g$, where $r_g \in R$. The addition and multiplication in $R[G]$ are defined as follows:

$$\begin{aligned} \sum_{g \in G} r_g g + \sum_{g \in G} s_g g &= \sum_{g \in G} (r_g + s_g) g, \\ \left(\sum_{g \in G} r_g g \right) \left(\sum_{h \in G} s_h h \right) &= \sum_{k \in G} \left(\sum_{g, h \in G, gh=k} r_g s_h \right) k. \end{aligned}$$

Building on the definitions above, we can now state the main theorem of [5], which provides a positive result for the Alon–Jaeger–Tarsi conjecture in the case of sufficiently large prime fields.

Theorem 3.6 (Nagy, Pach, 2025). *Let p be a prime such that $61 < p \neq 79$. If A is an $n \times n$ nonsingular matrix over $\mathbb{F}_p$ and $c_i, d_i \in \mathbb{F}_p$ ($1 \leq i \leq n$), then there exists a vector $x \in \mathbb{F}_p^n$ such that $x_i \neq c_i$ for each $1 \leq i \leq n$ and $(Ax)_i \neq d_i$ for each $1 \leq i \leq n$.*

Remark. Taking $c_i = d_i = 0$ gives the Alon–Jaeger–Tarsi conclusion for $p > 61$, $p \neq 79$.

We now summarize the main ingredients in Nagy and Pach’s proof. The argument can be organized into four steps.

1. **Characterization of counterexamples.** Nagy and Pach first introduce a hierarchy of properties (P1)–(P5) satisfied by a counterexample to Conjecture 1. In particular, they translate the nonexistence of a nowhere-zero vector into polynomial and group-ring identities.
2. **Additive-combinatorial estimates.** They establish estimates for the extremal quantities $S_1(p)$ and $N_1(p)$. For $p > 61$, $p \neq 79$, these estimates, together with explicit constructions for the remaining finite range, yield

$$S_1(p)N_1(p) < p - 1.$$

3. **Basis separation.** Their basis-separation lemma shows that, whenever

$$S_k(p)N_k(p) < p - 1,$$

suitable S_k -type and N_k -type subsets of $\mathbb{F}_p^\times$ can be chosen so that the corresponding coordinate sums with respect to two bases of $\mathbb{F}_p^n$ are disjoint.

4. **Dimension reduction.** Using the basis-separation lemma, Nagy and Pach reduce the relevant group-ring identity in dimension n to an analogous identity in dimension $n - 1$. Iterating this argument eventually reaches dimension 1, where the identity cannot hold, giving the desired contradiction.

For the notation used below, regard

$$G = \mathbb{F}_p^n$$

as an additive group. For $v \in G$, let g_v denote the corresponding basis element of the group ring. Thus

$$g_u g_v = g_{u+v}, \quad g_0 = 1.$$

Let $e_1, \dots, e_n$ denote the standard basis vectors of $\mathbb{F}_p^n$.

Lemma 3.7 (Equivalence and implication of the properties). *Let p be a prime, $n \geq 1$ be an integer and let A be a nonsingular $n \times n$ matrix over $\mathbb{F}_p$. Denote by $\omega = e^{\frac{2\pi i}{p}}$. For each $i \in \{1, 2, \dots, n\}$, let t_i, t'_i be integers satisfying $0 \leq t_i, t'_i \leq p - 1$. Let*

$$c_{i,1}, \dots, c_{i,t_i} \in \mathbb{F}_p$$

be distinct, and let

$$d_{i,1}, \dots, d_{i,t'_i} \in \mathbb{F}_p$$

be distinct. Let $a_1, \dots, a_n$ denote the rows of A . Then the following three statements are equivalent:

(P1) *There does not exist a vector $x \in \mathbb{F}_p^n$ such that $x_i \neq c_{i,k}$ for each $1 \leq i \leq n$, $1 \leq k \leq t_i$ and $(Ax)_i \neq d_{i,k}$ for each $1 \leq i \leq n$, $1 \leq k \leq t'_i$.*

(P2) *The polynomial*

$$h(x_1, \dots, x_n) = \prod_{\substack{1 \leq i \leq n, \\ 1 \leq k \leq t'_i}} \left(\sum_{1 \leq j \leq n} a_{i,j} x_j - d_{i,k} \right) \cdot \prod_{\substack{1 \leq i \leq n, \\ 1 \leq k \leq t_i}} (x_i - c_{i,k})$$

vanishes at every point $z = (z_1, \dots, z_n) \in \mathbb{F}_p^n$.

(P3) The element

$$\prod_{\substack{1 \leq i \leq n, \\ 1 \leq k \leq t_i}} (1 - \omega^{-c_{i,k}} g_{e_i}) \cdot \prod_{\substack{1 \leq i \leq n, \\ 1 \leq k \leq t'_i}} (1 - \omega^{-d_{i,k}} g_{a_i}) \in \mathbb{C}[G]$$

is zero.

The following statement follows from the equivalent statements (P1)-(P2)-(P3):

(P4) The element

$$\prod_{1 \leq i \leq n} (1 - g_{e_i})^{t_i} \cdot \prod_{1 \leq i \leq n} (1 - g_{a_i})^{t'_i} \in \mathbb{F}_p[G]$$

is zero.

The following statement follows from statement (P4):

(P5) For the polynomial f defined as

$$f(x_1, \dots, x_n) = \prod_{1 \leq i \leq n} \left(\sum_{1 \leq j \leq n} a_{i,j} x_j \right)^{t'_i} \cdot \prod_{1 \leq i \leq n} x_i^{t_i}$$

we have $\deg(\bar{f}) < \sum_{1 \leq i \leq n} t_i + \sum_{1 \leq i \leq n} t'_i$, where $\bar{f}$ is the reduction of f modulo $x_i^p - x_i$ for each $i \in \{1, 2, \dots, n\}$.

That is, $(P1) \Leftrightarrow (P2) \Leftrightarrow (P3) \Rightarrow (P4) \Rightarrow (P5)$. In particular, take $t_i = t'_i = 1$ and $c_{i,1} = d_{i,1} = 0$ for all $i \in \{1, 2, \dots, n\}$, then (P1) is exactly the negation of the Alon-Jaeger-Tarsi conjecture.

Proof. The equivalence of (P1) and (P2) is immediate: for $z \in \mathbb{F}_p^n$, we have $h(z) = 0$ if and only if at least one of the forbidden equalities

$$z_i = c_{i,k} \quad \text{or} \quad (Az)_i = d_{i,k}$$

holds.

We next prove $(P1) \Leftrightarrow (P3)$. For $z \in \mathbb{F}_p^n$, define the character

$$\chi_z : G \longrightarrow \mathbb{C}^\times, \quad \chi_z(g_v) = \omega^{\langle z, v \rangle},$$

which can be extended to $\tilde{\chi}_z : \mathbb{C}[G] \longrightarrow \mathbb{C}$ by linear combination. The Fourier transform on the finite abelian group G is injective. Hence an element of $\mathbb{C}[G]$ is zero if and only if all of its character evaluations are zero. Let U denote the group-ring element in (P3). Then

$$\tilde{\chi}_z(U) = \prod_{\substack{1 \leq i \leq n, \\ 1 \leq k \leq t_i}} (1 - \omega^{z_i - c_{i,k}}) \cdot \prod_{\substack{1 \leq i \leq n, \\ 1 \leq k \leq t'_i}} (1 - \omega^{\langle z, a_i \rangle - d_{i,k}}).$$

Since a_i is the i -th row of A , $\langle z, a_i \rangle = (Az)_i$. Because ω is a primitive p -th root of unity, $\tilde{\chi}_z(U) = 0$ if and only if at least one of the equalities

$$z_i = c_{i,k}, \quad (Az)_i = d_{i,k}$$

holds. Thus $U = 0$ if and only if (P1) holds.

We now prove $(P3) \Rightarrow (P4)$. Expand the element in (P3). For each $v \in G$, its coefficient can be written in the form

$$\sum_{s=0}^{p-1} m_{v,s} \omega^s, \quad m_{v,s} \in \mathbb{Z}.$$

If (P3) holds, then

$$\sum_{s=0}^{p-1} m_{v,s} \omega^s = 0.$$

The minimal polynomial of ω over $\mathbb{Q}$ is $\Phi_p(X) = 1 + X + \cdots + X^{p-1}$. Since the polynomial $\sum_{s=0}^{p-1} m_{v,s} X^s$ has degree at most $p-1$ and vanishes at ω , it is an integer multiple of Φ_p . Hence $m_{v,0} = m_{v,1} = \cdots = m_{v,p-1}$. Consequently,

$$\sum_{s=0}^{p-1} m_{v,s} \equiv 0 \pmod{p}.$$

If all powers of ω in the expansion are replaced by 1, the coefficient of g_v becomes precisely

$$\sum_{s=0}^{p-1} m_{v,s}.$$

Therefore every coefficient of

$$\prod_{i=1}^n (1 - g_{e_i})^{t_i} \prod_{i=1}^n (1 - g_{a_i})^{t'_i}$$

is divisible by p , proving (P4).

Finally, we prove (P4) $\implies$ (P5). Put $w_i = 1 - g_{e_i}$, $1 \leq i \leq n$. Since G has exponent p ,

$$w_i^p = (1 - g_{e_i})^p = 1 - g_{pe_i} = 0.$$

Moreover,

$$\left\{ \prod_{i=1}^n w_i^{b_i} \mid 0 \leq b_i \leq p-1 \right\}$$

is a basis of $\mathbb{F}_p[G]$ over $\mathbb{F}_p$. Indeed, $g_{e_i} = 1 - w_i$, so these monomials span $\mathbb{F}_p[G]$; their number is $p^n = \dim_{\mathbb{F}_p} \mathbb{F}_p[G]$.

Write

$$a_i = \sum_{j=1}^n a_{ij} e_j.$$

Then

$$g_{a_i} = \prod_{j=1}^n (1 - w_j)^{a_{ij}},$$

where the entries a_{ij} are represented by integers in $\{0, \dots, p-1\}$. Hence

$$1 - g_{a_i} = \sum_{j=1}^n a_{ij} w_j + \text{terms of total degree at least 2}.$$

Set

$$D = \sum_{i=1}^n t_i + \sum_{i=1}^n t'_i.$$

The homogeneous component of total degree D in

$$\prod_{i=1}^n w_i^{t_i} \prod_{i=1}^n (1 - g_{a_i})^{t'_i}$$

is exactly

$$\prod_{i=1}^n w_i^{t_i} \prod_{i=1}^n \left(\sum_{j=1}^n a_{ij} w_j \right)^{t'_i},$$

which is obtained from f by replacing x_i with w_i .

By (P4), the group-ring element is zero. Therefore every nonzero degree- D monomial occurring in f must contain some variable with exponent at least p ; otherwise the corresponding monomial in the basis above would survive with nonzero coefficient. Reducing such a monomial modulo $x_i^p - x_i$ strictly decreases its total degree. Hence

$$\deg \bar{f} < D,$$

which proves (P5). □

Lemma 3.8 (Arithmetic-combinatorial estimate). *Let $p > 3$ be a prime. Then*

$$S_1(p) = N_1(p) \leq 2\lfloor \log_2 p \rfloor + 2.$$

More precisely, if p is not of the form $p = 2^r - 1$ for some integer $r \geq 3$, then the sharper estimate

$$S_1(p) = N_1(p) \leq 2\lfloor \log_2 p \rfloor$$

holds.

Proof. We first show that $S_1(p) = N_1(p)$. By definition, every N_1 -type set is S_1 -type, so $S_1(p) \leq N_1(p)$. Conversely, let $K \subseteq \mathbb{F}_p$ be an S_1 -type set and let $b \in \mathbb{F}_p \setminus K$. Choose any $a \in K$ and put $d = a - b$. Since $a \neq b$, we have $d \in \mathbb{F}_p^\times$, and $b + d = a \in K$. Thus every S_1 -type set is also N_1 -type. Hence

$$S_1(p) = N_1(p).$$

It therefore remains to construct a sufficiently small S_1 -type set. We distinguish two cases.

Case 1: $p \equiv 1 \pmod{4}$.

Write $p = 4m + 1$. Let

$$L = \lfloor \log_2 m \rfloor \quad \text{and} \quad r_j = \left\lfloor \frac{m}{2^j} \right\rfloor, \quad 0 \leq j \leq L.$$

In particular, $r_0 = m$, $r_L = 1$. Set

$$K = \left\{ 0, \frac{p-1}{2} \right\} \cup \{ \pm r_j \mid 0 \leq j \leq L \}.$$

We verify that K is S_1 -type. First,

$$2 \cdot 0 = 1 + (-1), \quad 2 \cdot \frac{p-1}{2} = (-1) + 0 \quad \text{in } \mathbb{F}_p.$$

Moreover,

$$\begin{aligned} 2r_0 &= 2m = \frac{p-1}{2} + 0, \\ 2(-r_0) &= -2m = \frac{p-1}{2} + 1 \quad \text{in } \mathbb{F}_p. \end{aligned}$$

For $1 \leq j \leq L$, if r_{j-1} is even, then

$$\begin{aligned} 2r_j &= r_{j-1} + 0, \\ 2(-r_j) &= (-r_{j-1}) + 0. \end{aligned}$$

If r_{j-1} is odd, then

$$\begin{aligned} 2r_j &= r_{j-1} + (-1), \\ 2(-r_j) &= (-r_{j-1}) + 1. \end{aligned}$$

Thus every element of K is the middle term of a nontrivial three-term arithmetic progression contained in K .

Therefore K is S_1 -type, and $|K| \leq 2(L+1) + 2$. Since $2^L \leq m < 2^{L+1}$ and $p = 4m + 1$, we have

$$\lfloor \log_2 p \rfloor = L + 2.$$

Consequently,

$$|K| \leq 2(L+1) + 2 = 2\lfloor \log_2 p \rfloor.$$

Case 2: $p \equiv 3 \pmod{4}$.

Write $p = 4m + 3$. Let

$$L = \lfloor \log_2(m+1) \rfloor \quad \text{and} \quad r_j = \left\lfloor \frac{m+1}{2^j} \right\rfloor, \quad 0 \leq j \leq L.$$

Then $r_0 = m + 1$, $r_L = 1$. Set

$$K = \left\{0, \frac{p-1}{2}\right\} \cup \{\pm r_j \mid 0 \leq j \leq L\}.$$

Again,

$$2 \cdot 0 = 1 + (-1), \quad 2 \cdot \frac{p-1}{2} = (-1) + 0 \quad \text{in } \mathbb{F}_p.$$

Moreover,

$$\begin{aligned} 2r_0 &= 2(m+1) = \frac{p-1}{2} + 1, \\ 2(-r_0) &= -2(m+1) = \frac{p-1}{2} = \frac{p-1}{2} + 0 \quad \text{in } \mathbb{F}_p. \end{aligned}$$

For $j \geq 1$, exactly the same parity argument as in Case 1 shows that both r_j and $-r_j$ are middle terms of three-term arithmetic progressions contained in K . Hence K is S_1 -type. Its size satisfies

$$|K| \leq 2(L+1) + 2 = 2L + 4.$$

If $m+1$ is not a power of 2, then $2^L < m+1 < 2^{L+1}$. Since $p = 4(m+1) - 1$, we obtain

$$2^{L+2} - 1 < p < 2^{L+3} - 1,$$

and hence

$$\lfloor \log_2 p \rfloor = L + 2.$$

Therefore

$$|K| \leq 2L + 4 = 2\lfloor \log_2 p \rfloor.$$

If $m+1$ is a power of 2, say $m+1 = 2^L$, then $p = 4(m+1) - 1 = 2^{L+2} - 1$. Thus

$$\lfloor \log_2 p \rfloor = L + 1,$$

and consequently

$$|K| \leq 2L + 4 = 2\lfloor \log_2 p \rfloor + 2.$$

The exceptional situation occurs precisely when p is of the form $2^r - 1$. Combining the two cases proves

$$S_1(p) = N_1(p) \leq 2\lfloor \log_2 p \rfloor + 2,$$

with the sharper bound

$$S_1(p) = N_1(p) \leq 2\lfloor \log_2 p \rfloor$$

whenever p is not of this form. □

Lemma 3.9 (Estimate for $S_1(p)N_1(p)$). *For $p > 61$, $p \neq 79$, we have*

$$S_1(p)N_1(p) < p - 1.$$

Proof. By Lemma 3.8, $S_1(p) = N_1(p)$, so it suffices to prove

$$S_1(p)^2 < p - 1.$$

For the primes $61 < p < 199$, $p \neq 79$, and for $p = 257$, Nagy and Pach give explicit S_1 -type subsets whose cardinality is strictly smaller than $\sqrt{p-1}$. Hence $S_1(p)^2 < p - 1$ for all of these primes.

p	construction	size
67	0, 1, 3, 6, 11, 35, 54, 66	8
71	0, 1, 3, 14, 36, 55, 63, 70	8
73	0, 1, 3, 6, 20, 37, 65, 72	8
83	0, 1, 2, 3, 7, 14, 44, 67, 82	9
89	0, 1, 2, 4, 8, 15, 47, 72, 88	9
97	0, 1, 2, 4, 8, 26, 51, 87, 96	9
101	0, 1, 2, 5, 9, 18, 56, 81, 100	9
103	0, 1, 2, 5, 10, 19, 59, 82, 102	9
107	0, 1, 2, 4, 9, 29, 56, 96, 106	9
109	0, 1, 2, 5, 9, 29, 58, 98, 108	9
113	0, 1, 2, 5, 17, 29, 58, 94, 112	9
127	0, 1, 2, 5, 17, 65, 96, 120, 126	9
131	0, 1, 3, 6, 11, 35, 70, 118, 130	9
137	0, 1, 3, 6, 37, 62, 73, 87, 136	9
139	0, 1, 3, 6, 12, 38, 73, 125, 138	9
149	0, 1, 3, 6, 20, 76, 113, 141, 148	9
151	0, 1, 3, 6, 20, 76, 115, 143, 150	9
157	0, 1, 3, 6, 9, 15, 29, 89, 126, 156	10
163	0, 1, 3, 6, 9, 17, 28, 86, 133, 162	10
167	0, 1, 3, 6, 9, 17, 31, 95, 134, 166	10
173	0, 1, 3, 6, 9, 17, 34, 59, 112, 172	10
179	0, 1, 3, 6, 9, 17, 34, 62, 115, 178	10
181	0, 1, 3, 6, 9, 17, 81, 134, 162, 180	10
191	0, 1, 3, 6, 9, 17, 53, 100, 172, 190	10
193	0, 1, 3, 6, 9, 17, 34, 104, 157, 192	10
197	0, 1, 3, 6, 9, 17, 53, 106, 178, 196	10
257	0, 1, 3, 6, 9, 18, 35, 61, 87, 168, 256	11

It remains to consider

$$p \geq 199, \quad p \neq 257.$$

First suppose that p is not of the form $2^s - 1$. Put $r = \lfloor \log_2 p \rfloor$. Lemma 3.8 gives $S_1(p) \leq 2r$.

If $199 \leq p \leq 255$, then $r = 7$, and therefore

$$S_1(p)^2 \leq 4 \cdot 7^2 = 196 < p - 1.$$

If $256 < p < 512$, then, since p is prime and $p \neq 257$, we have $p \geq 263$. Here $r = 8$, so

$$S_1(p)^2 \leq 4 \cdot 8^2 = 256 < p - 1.$$

Finally, suppose $p \geq 512$. Then $r \geq 9$ and $p \geq 2^r$. For every integer $r \geq 9$,

$$4r^2 < 2^r - 1.$$

Indeed, this is true for $r = 9$, since

$$4 \cdot 9^2 = 324 < 511,$$

and the inequality is preserved when r is increased. Consequently,

$$S_1(p)^2 \leq 4r^2 < 2^r - 1 \leq p - 1.$$

It remains only to consider the case in which $p = 2^s - 1$. Since $p \geq 199$ and p is prime, we have $s \geq 9$. The second bound in Lemma 3.8 gives

$$S_1(p) \leq 2\lfloor \log_2 p \rfloor + 2 = 2s.$$

Hence

$$S_1(p)^2 \leq 4s^2.$$

For every $s \geq 9$,

$$4s^2 < 2^s - 2,$$

and therefore

$$S_1(p)^2 < 2^s - 2 = p - 1.$$

Thus in every case $S_1(p)^2 < p - 1$. □

Lemma 3.10 (Basis-separation). *Assume $S_k(p)N_k(p) < p - 1$. For any two bases $(e_i)_{i=1}^n$ and $(f_i)_{i=1}^n$ of $\mathbb{F}_p^n$, there exist $A_i, B_i \subseteq \mathbb{F}_p^\times$, $1 \leq i \leq n$ such that A_i is S_k -type, B_i is N_k -type, and*

$$\sum_{i=1}^n x_i e_i \neq \sum_{i=1}^n y_i f_i \quad \text{whenever } x_i \in A_i, y_i \in B_i.$$

Proof. Choose an S_k -type set U and an N_k -type set V such that

$$|U| = S_k(p), \quad |V| = N_k(p).$$

The S_k - and N_k -properties are invariant under translations and nonzero dilations. Since $S_k(p)N_k(p) < p - 1$, both sets have size strictly less than p , so after translating them if necessary we may assume

$$U, V \subseteq \mathbb{F}_p^\times.$$

Choose $\lambda_1, \dots, \lambda_n \in \mathbb{F}_p^\times$ independently and uniformly at random. Fix

$$x_i \in U, \quad y_i \in V.$$

Consider the equation

$$\sum_{i=1}^n x_i e_i = \sum_{i=1}^n \lambda_i y_i f_i.$$

Writing the left-hand side in the basis (f_i) determines at most one possible value of each λ_i , since $y_i \neq 0$. Therefore

$$\mathbb{P} \left(\sum_{i=1}^n x_i e_i = \sum_{i=1}^n \lambda_i y_i f_i \right) \leq \frac{1}{(p-1)^n}.$$

By the union bound, the probability that this equality holds for at least one choice of the x_i and y_i is at most

$$\frac{|U|^n |V|^n}{(p-1)^n} = \left(\frac{S_k(p)N_k(p)}{p-1} \right)^n < 1.$$

Hence there exists a choice of $\lambda_1, \dots, \lambda_n$ for which no such equality occurs.

Set $A_i = U$, $B_i = \lambda_i V$. Nonzero dilation preserves the N_k -property, so the sets A_i, B_i have all the required properties. □

Lemma 3.11 (Auxiliary coefficient criterion). *Assume $k < p$. Let A be nonsingular with row basis $a_1, \dots, a_n$, and let $\lambda_i, \mu_i \in \mathbb{F}_p^\times$. Let*

$$\alpha = (\alpha_1, \dots, \alpha_n), \quad \beta = (\beta_1, \dots, \beta_n) \in \{0, \dots, k\}^n.$$

For $r = (r_1, \dots, r_n)$ with $0 \leq r_i \leq \alpha_i$, define $F_r^\alpha : \mathbb{F}_p^n \rightarrow \mathbb{F}_p$ by

$$F_r^\alpha \left(\sum_{i=1}^n (\ell_i - r_i) \lambda_i e_i \right) = (-1)^{\sum_i \ell_i} \prod_{i=1}^n \binom{\alpha_i}{\ell_i}, \quad 0 \leq \ell_i \leq \alpha_i,$$

and let F_r^α be zero elsewhere. Similarly, for $s = (s_1, \dots, s_n)$ with $0 \leq s_i \leq \beta_i$, define $H_s^\beta : \mathbb{F}_p^n \rightarrow \mathbb{F}_p$ by

$$H_s^\beta \left(\sum_{i=1}^n (s_i - m_i) \mu_i a_i \right) = (-1)^{\sum_i m_i} \prod_{i=1}^n \binom{\beta_i}{m_i}, \quad 0 \leq m_i \leq \beta_i,$$

and let H_s^β be zero elsewhere. Since $\alpha_i, \beta_i \leq k < p$ and both $(e_i)_{i=1}^n$ and $(a_i)_{i=1}^n$ are bases, these definitions are unambiguous.

Then

$$\prod_{i=1}^n (1 - g_{e_i})^{\alpha_i} \prod_{i=1}^n (1 - g_{a_i})^{\beta_i} = 0$$

if and only if

$$\langle F_r^\alpha, H_s^\beta \rangle := \sum_{v \in \mathbb{F}_p^n} F_r^\alpha(v) H_s^\beta(v) = 0$$

for every r, s satisfying $0 \leq r_i \leq \alpha_i$ and $0 \leq s_i \leq \beta_i$.

Proof. Let

$$Q_{\alpha, \beta} = \prod_{i=1}^n (1 - g_{e_i})^{\alpha_i} \prod_{i=1}^n (1 - g_{a_i})^{\beta_i}$$

and consider the scaled product

$$P_{\lambda, \mu} = \prod_{i=1}^n (1 - g_{\lambda_i e_i})^{\alpha_i} \prod_{i=1}^n (1 - g_{\mu_i a_i})^{\beta_i}.$$

We first show that

$$P_{\lambda, \mu} = 0 \iff Q_{\alpha, \beta} = 0.$$

Let $c \in \mathbb{F}_p^\times$ and let $v \in G$ be nonzero. Choose the integer representative $\tilde{c} \in \{1, \dots, p-1\}$ of c . Since G is written additively,

$$g_{cv} = g_v^{\tilde{c}}.$$

Hence

$$1 - g_{cv} = 1 - g_v^{\tilde{c}} = (1 - g_v) (1 + g_v + \dots + g_v^{\tilde{c}-1}).$$

Set $u_{c,v} = 1 + g_v + \dots + g_v^{\tilde{c}-1}$. We claim that $u_{c,v}$ is a unit in $\mathbb{F}_p[G]$.

Recall the augmentation map

$$\varepsilon : \mathbb{F}_p[G] \longrightarrow \mathbb{F}_p, \quad \varepsilon \left(\sum_{g \in G} \alpha_g g \right) = \sum_{g \in G} \alpha_g,$$

and let $I = \ker \varepsilon$ be the augmentation ideal. Since

$$G \cong (\mathbb{Z}/p\mathbb{Z})^n,$$

the group algebra may be identified with

$$\mathbb{F}_p[G] \cong \frac{\mathbb{F}_p[w_1, \dots, w_n]}{(w_1^p, \dots, w_n^p)}, \quad w_i = 1 - g_{e_i}.$$

In particular, the augmentation ideal $I = (w_1, \dots, w_n)$ is nilpotent.

Now

$$\varepsilon(u_{c,v}) = \tilde{c} = c \neq 0 \quad \text{in } \mathbb{F}_p.$$

Thus we may write

$$u_{c,v} = c(1 + x)$$

for some $x \in I$. Since I is nilpotent, there exists N such that $x^N = 0$, and therefore

$$(1 + x)^{-1} = 1 - x + x^2 - \dots + (-x)^{N-1}.$$

It follows that

$$u_{c,v}^{-1} = c^{-1} (1 - x + x^2 - \dots + (-x)^{N-1}),$$

so $u_{c,v}$ is indeed a unit.

Consequently,

$$1 - g_{cv} = (1 - g_v) u_{c,v}$$

differs from $1 - g_v$ only by multiplication by a unit.

Applying this to every λ_i and μ_i , we obtain units $u_i, v_i \in \mathbb{F}_p[G]$ such that

$$1 - g_{\lambda_i e_i} = (1 - g_{e_i})u_i, \quad 1 - g_{\mu_i a_i} = (1 - g_{a_i})v_i.$$

Therefore

$$P_{\lambda, \mu} = Q_{\alpha, \beta} \left(\prod_{i=1}^n u_i^{\alpha_i} \right) \left(\prod_{i=1}^n v_i^{\beta_i} \right).$$

The second factor is a unit, and hence

$$P_{\lambda, \mu} = 0 \iff Q_{\alpha, \beta} = 0.$$

It remains to characterize the vanishing of $P_{\lambda, \mu}$ in terms of the functions F_r^α and H_s^β . Expanding the binomial factors gives

$$P_{\lambda, \mu} = \sum_{\ell_1=0}^{\alpha_1} \cdots \sum_{\ell_n=0}^{\alpha_n} \sum_{m_1=0}^{\beta_1} \cdots \sum_{m_n=0}^{\beta_n} (-1)^{\sum_i \ell_i + \sum_i m_i} \left(\prod_{i=1}^n \binom{\alpha_i}{\ell_i} \binom{\beta_i}{m_i} \right) \\ \cdot g_{\sum_{i=1}^n \ell_i \lambda_i e_i + \sum_{i=1}^n m_i \mu_i a_i}.$$

Fix r, s with $0 \leq r_i \leq \alpha_i$ and $0 \leq s_i \leq \beta_i$, and put

$$\xi_{r, s} = \sum_{i=1}^n r_i \lambda_i e_i + \sum_{i=1}^n s_i \mu_i a_i.$$

A term indexed by $(\ell_1, \dots, \ell_n)$ and $(m_1, \dots, m_n)$ contributes to the coefficient of $g_{\xi_{r, s}}$ precisely when

$$\sum_{i=1}^n \ell_i \lambda_i e_i + \sum_{i=1}^n m_i \mu_i a_i = \sum_{i=1}^n r_i \lambda_i e_i + \sum_{i=1}^n s_i \mu_i a_i.$$

Equivalently,

$$\sum_{i=1}^n (\ell_i - r_i) \lambda_i e_i = \sum_{i=1}^n (s_i - m_i) \mu_i a_i.$$

Let the common vector in the last equality be v . By the definitions of F_r^α and H_s^β , the contribution of this pair (ℓ, m) is exactly

$$F_r^\alpha(v) H_s^\beta(v).$$

Summing over all possible common vectors $v \in \mathbb{F}_p^n$, we obtain

$$[g_{\xi_{r, s}}] P_{\lambda, \mu} = \sum_{v \in \mathbb{F}_p^n} F_r^\alpha(v) H_s^\beta(v) = \langle F_r^\alpha, H_s^\beta \rangle,$$

where $[g_{\xi_{r, s}}] P_{\lambda, \mu}$ denotes the coefficient of $g_{\xi_{r, s}}$ in $P_{\lambda, \mu}$.

Thus, if $P_{\lambda, \mu} = 0$, then every coefficient vanishes, and therefore $\langle F_r^\alpha, H_s^\beta \rangle = 0$ for every admissible r, s . Conversely, suppose that $\langle F_r^\alpha, H_s^\beta \rangle = 0$ for every admissible r, s . Every group element appearing in the expansion of $P_{\lambda, \mu}$ has the form

$$\sum_{i=1}^n r_i \lambda_i e_i + \sum_{i=1}^n s_i \mu_i a_i$$

for some admissible r, s ; indeed, we may simply take $r_i = \ell_i$ and $s_i = m_i$ for one term contributing to that group element. Hence every coefficient of $P_{\lambda, \mu}$ is equal to $\langle F_r^\alpha, H_s^\beta \rangle$ for some admissible r, s , and therefore every coefficient is zero. Thus

$$P_{\lambda, \mu} = 0.$$

Combining this with

$$P_{\lambda, \mu} = 0 \iff Q_{\alpha, \beta} = 0$$

proves the result. □

Lemma 3.12 (Group-ring dimension-reduction). *Assume $S_k(p)N_k(p) < p - 1$. If a nonsingular matrix satisfies*

$$\prod_{i=1}^n (1 - g_{e_i})^k \prod_{i=1}^n (1 - g_{a_i})^k = 0 \quad \text{in } \mathbb{F}_p[G],$$

then the same type of identity holds for a nonsingular $(n - 1) \times (n - 1)$ matrix.

Proof. By Lemma 3.10, applied to the bases $(e_i)_{i=1}^n$ and $(a_i)_{i=1}^n$, choose S_k -type sets $A_i \subseteq \mathbb{F}_p^\times$ and N_k -type sets $B_i \subseteq \mathbb{F}_p^\times$ such that

$$\sum_{i=1}^n x_i e_i \neq \sum_{i=1}^n y_i a_i$$

whenever $x_i \in A_i$, $y_i \in B_i$.

Define

$$\mathcal{S} = \left\{ \sum_{i=1}^n y_i e_i \mid y_i \in A_i \right\}.$$

For $v \in \mathcal{S}$, write uniquely

$$v = \sum_{i=1}^n z_{v,i} a_i,$$

and define $I_v = \{1 \leq i \leq n \mid z_{v,i} \in B_i\}$. By the separation property, $I_v \neq \{1, \dots, n\}$ for every $v \in \mathcal{S}$.

Choose $u \in \mathcal{S}$ such that I_u is maximal under inclusion, and choose $j \notin I_u$. Write

$$u = \sum_{i=1}^n y_{u,i} e_i = \sum_{i=1}^n z_{u,i} a_i.$$

Since each A_i is S_k -type, there exists $\lambda_i \in \mathbb{F}_p^\times$ such that

$$y_{u,i} + t\lambda_i \in A_i \quad (-k \leq t \leq k).$$

For $i \in I_u$, since B_i is N_k -type and hence S_k -type, choose $\mu_i \in \mathbb{F}_p^\times$ such that

$$z_{u,i} + t\mu_i \in B_i \quad (-k \leq t \leq k).$$

For $i \notin I_u$ and $i \neq j$, choose μ_i arbitrarily in $\mathbb{F}_p^\times$.

Finally, because $z_{u,j} \notin B_j$ and B_j is N_k -type, there exists $\mu_j \in \mathbb{F}_p^\times$ such that

$$z_{u,j} + t\mu_j \in B_j \quad (1 \leq t \leq k).$$

We first claim that one of the a_i -factors can be deleted while preserving the zero identity:

$$\prod_{i=1}^n (1 - g_{e_i})^k \prod_{\substack{i=1 \\ i \neq j}}^n (1 - g_{a_i})^k = 0. \quad (\star)$$

Set $\alpha = (k, \dots, k)$ and $\beta = (k, \dots, k)$. By Lemma 3.11, the assumed zero identity implies

$$\langle F_r^\alpha, H_s^\beta \rangle = 0$$

for every $r, s \in \{0, \dots, k\}^n$.

Fix arbitrary

$$r_i \in \{0, \dots, k\} \quad (1 \leq i \leq n)$$

and

$$s_i \in \{0, \dots, k\} \quad (i \neq j),$$

and extend s by setting $s_j = k$. Translate both functions appearing in the coefficient criterion by u . Translation does not change their inner product.

Every point in the translated support on the e -side has the form

$$u + \sum_{i=1}^n (\ell_i - r_i) \lambda_i e_i = \sum_{i=1}^n (y_{u,i} + (\ell_i - r_i) \lambda_i) e_i.$$

Since $-k \leq \ell_i - r_i \leq k$, all its e_i -coordinates belong to A_i . Thus every such point lies in $\mathcal{S}$.

On the a -side, a point has the form

$$u + \sum_{i=1}^n (s_i - m_i) \mu_i a_i = \sum_{i=1}^n (z_{u,i} + (s_i - m_i) \mu_i) a_i.$$

If $i \in I_u$, the corresponding coefficient remains in B_i , because $-k \leq s_i - m_i \leq k$. Moreover, if $m_j < k$, then $1 \leq k - m_j \leq k$, and hence

$$z_{u,j} + (k - m_j) \mu_j \in B_j.$$

Therefore, if a point with $m_j < k$ belonged to both translated supports, it would give some $w \in \mathcal{S}$ satisfying

$$I_w \supseteq I_u \cup \{j\},$$

contradicting the maximality of I_u . Hence all terms with $m_j < k$ contribute zero to the inner product.

The only possible contribution has $m_j = k$. Its j -th factor is

$$(-1)^k \binom{k}{k} = (-1)^k,$$

and its displacement in the a_j direction is zero. Let

$$\beta_i^{(j)} = \begin{cases} k, & i \neq j, \\ 0, & i = j, \end{cases} \quad s'_i = \begin{cases} s_i, & i \neq j, \\ 0, & i = j. \end{cases}$$

It follows that

$$0 = \langle F_r^\alpha, H_s^\beta \rangle = (-1)^k \langle F_r^\alpha, H_{s'}^{\beta^{(j)}} \rangle.$$

Since the choices of r_i and s_i for $i \neq j$ were arbitrary,

$$\langle F_r^\alpha, H_{s'}^{\beta^{(j)}} \rangle = 0$$

for every r with $0 \leq r_i \leq k$ and every s' with $0 \leq s'_i \leq \beta_i^{(j)}$. Applying Lemma 3.11 with $\alpha = (k, \dots, k)$ and $\beta = \beta^{(j)}$ gives $(\star)$.

We now pass from dimension n to dimension $n - 1$. Since A is nonsingular, expansion of $\det(A)$ along row j shows that there exists an index $\ell \in \{1, \dots, n\}$ such that the $(n - 1) \times (n - 1)$ minor obtained by deleting row j and column ℓ is nonsingular. Let

$$H = \text{span}_{\mathbb{F}_p} \{e_r \mid r \neq \ell\}.$$

For $i \neq j$, write $a_i = a'_i + a_{i\ell} e_\ell$, $a'_i \in H$. The vectors $\{a'_i \mid i \neq j\}$ form a basis of H , since their coordinate matrix is precisely the nonsingular minor chosen above.

Put $w = 1 - g_{e_\ell}$. For every $i \neq j$,

$$\begin{aligned} 1 - g_{a_i} &= 1 - g_{a'_i} g_{a_{i\ell} e_\ell} \\ &= (1 - g_{a'_i}) + g_{a'_i} (1 - g_{a_{i\ell} e_\ell}). \end{aligned}$$

Since $1 - g_{a_{i\ell} e_\ell}$ is divisible by $1 - g_{e_\ell} = w$, there exists $y_i \in \mathbb{F}_p[G]$ such that

$$1 - g_{a_i} = (1 - g_{a'_i}) + w y_i.$$

Substituting this into $(\star)$ gives

$$0 = w^k \prod_{\substack{r=1 \\ r \neq \ell}}^n (1 - g_{e_r})^k \prod_{\substack{i=1 \\ i \neq j}}^n ((1 - g_{a'_i}) + w y_i)^k.$$

Because

$$G = H \oplus \langle e_\ell \rangle,$$

the group algebra $\mathbb{F}_p[G]$ is a free $\mathbb{F}_p[H]$ -module with basis

$$1, w, w^2, \dots, w^{p-1}.$$

The coefficient of the lowest possible power w^k in the last display is

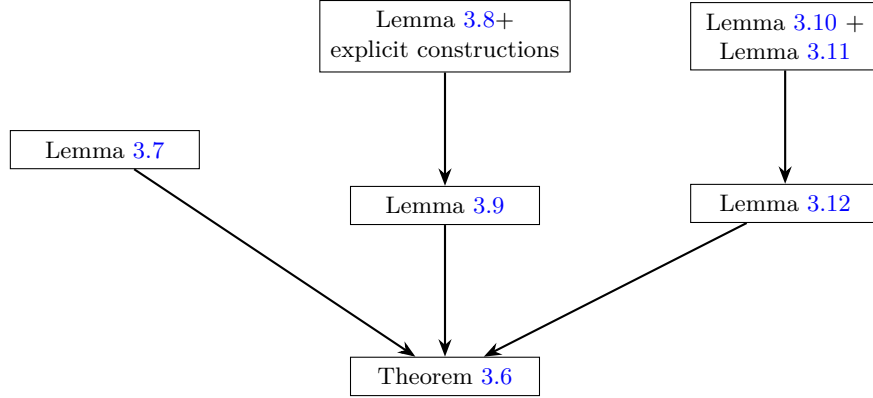
$$\prod_{\substack{r=1 \\ r \neq \ell}}^n (1 - g_{e_r})^k \prod_{\substack{i=1 \\ i \neq j}}^n (1 - g_{a'_i})^k.$$

Since the whole expression is zero, this coefficient must be zero. Therefore

$$\prod_{\substack{r=1 \\ r \neq \ell}}^n (1 - g_{e_r})^k \prod_{\substack{i=1 \\ i \neq j}}^n (1 - g_{a'_i})^k = 0 \quad \text{in } \mathbb{F}_p[H].$$

Identifying H with $\mathbb{F}_p^{n-1}$, this is precisely the same group-ring identity for the nonsingular $(n-1) \times (n-1)$ minor of A . $\square$

The quoted results combine to yield Theorem 3.6, as summarized in the following diagram:



Proof of Theorem 3.6. Suppose, for contradiction, that the conclusion fails. Thus there exist

$$c_i, d_i \in \mathbb{F}_p \quad 1 \leq i \leq n$$

such that no $x \in \mathbb{F}_p^n$ satisfies simultaneously

$$x_i \neq c_i \quad \text{and} \quad (Ax)_i \neq d_i$$

for all $1 \leq i \leq n$.

Apply Lemma 3.7 with

$$t_i = t'_i = 1, \quad c_{i,1} = c_i, \quad d_{i,1} = d_i.$$

Property (P1) holds, and hence (P4) holds:

$$\prod_{i=1}^n (1 - g_{e_i}) \prod_{i=1}^n (1 - g_{a_i}) = 0 \quad \text{in } \mathbb{F}_p[\mathbb{F}_p^n]. \quad (\dagger)$$

By Lemma 3.9,

$$S_1(p)N_1(p) < p - 1.$$

Therefore Lemma 3.12, with $k = 1$, can be applied repeatedly to $(\dagger)$. After $n-1$ applications we obtain a nonsingular 1×1 matrix for which the analogous identity holds.

Let its nonzero entry be $c \in \mathbb{F}_p^\times$. In dimension 1, with basis vector e , the identity would be

$$(1 - g_e)(1 - g_{ce}) = 0.$$

But $1 - g_{ce} = (1 - g_e)u$ for some unit $u \in \mathbb{F}_p[\mathbb{F}_p]$. Hence the left-hand side differs from $(1 - g_e)^2$ by multiplication by a unit. If $w = 1 - g_e$, then

$$1, w, w^2, \dots, w^{p-1}$$

is a basis of $\mathbb{F}_p[\mathbb{F}_p]$. Since $p > 61 > 2$, $w^2 \neq 0$. This is a contradiction. Therefore the assumed counterexample cannot exist, proving Theorem 3.6. $\square$

4 Some Generalizations and Related Results

In this section, we discuss some generalizations and related results of the Alon–Jaeger–Tarsi conjecture.

4.1 A Permanent Analogue

Replacing the determinant in Conjecture 1 by the permanent leads to the following analogue, which was proved by Alon in [2].

Lemma 4.1 (Permanent Lemma). *Let $\mathbb{F}_q$ be a finite field with $q > 2$ and let A be an $n \times n$ matrix over $\mathbb{F}_q$ with $\text{Per}(A) \neq 0$. Suppose $S_1, S_2, \dots, S_n \subseteq \mathbb{F}_q$ with $|S_i| = 2$ and $b = (b_1, b_2, \dots, b_n) \in \mathbb{F}_q^n$. Then there exists a vector $x \in S_1 \times S_2 \times \dots \times S_n$ such that $(Ax)_i \neq b_i$ for all i .*

Proof. Consider the polynomial

$$f(x_1, x_2, \dots, x_n) = \prod_{i=1}^n (a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n - b_i).$$

Note that $f(x) \neq 0 \Leftrightarrow (Ax)_i \neq b_i$ for all i . So we only need to show that there exists a vector $x \in S_1 \times S_2 \times \dots \times S_n$ such that $f(x) \neq 0$. Since $|S_i| = 2$, $\deg(f) = n$, and the coefficient of $x_1x_2 \dots x_n$ in f is exactly $\text{Per}(A) \neq 0$, by the Combinatorial Nullstellensatz, there exists a vector $x \in S_1 \times S_2 \times \dots \times S_n$ such that $f(x) \neq 0$. $\square$

Corollary 4.2. *Let $\mathbb{F}_q$ be a finite field with $q > 2$ and let $A \in M_n(\mathbb{F}_q)$ satisfy $\text{Per}(A) \neq 0$. Then there exists a vector $x \in \mathbb{F}_q^n$ such that both x and Ax are nowhere-zero.*

4.2 The Permanent Conjecture

Motivated by Alon’s method, Kahn proposed the so-called *Permanent Conjecture*. To give the statement of this conjecture, we need to define the *permanent rank*.

Definition 4.3 (Permanent rank). The permanent rank of a matrix A is defined to be the size of the largest square submatrix with nonzero permanent, and is denoted by $\text{perrank}(A)$.

Conjecture 2 (Kahn). *The permanent rank of the matrix $(A \mid A)$ is n for every $n \times n$ nonsingular matrix A , where $(A \mid A)$ is the $n \times 2n$ matrix obtained by concatenating the columns.*

We show that Conjecture 2 implies Conjecture 1. If the permanent rank of $(A \mid A)$ is n , then there exists a $n \times n$ submatrix of $(A \mid A)$ with nonzero permanent. Recalling Alon’s argument, we can find a vector $\alpha \in L$ such that $\alpha_i \in \{0, 1, 2\}$ for all i and $\text{Per}(A_\alpha) \neq 0$. Since $2 < p - 1$, no other monomial reduces to the same exponent vector, so cancellation cannot occur, which implies that the polynomial Q_A is not identically zero over $(\mathbb{F}_p^\times)^n$. Hence we can find a vector x such that both x and Ax are nowhere-zero.

In fact, from the argument above, to prove Conjecture 1, it suffices to show that there exists a vector $\alpha \in L$ such that $\alpha_i \in \{0, 1, 2, 3\}$ for all i and $\text{Per}(A_\alpha) \neq 0$. Hence we can also consider the weaker version of Conjecture 2, i.e., the permanent rank of the matrix $(A \mid A \mid A)$ is n for every $n \times n$ nonsingular matrix A .

We recall the following results of Yu [6].

Lemma 4.4. *Suppose A is an $n \times m$ matrix with $\text{perrank}(A) = m \leq n$. Let $(A \mid x)$ be the $n \times (m + 1)$ matrix obtained by adjoining the vector x to A as its final column. Then*

$$V := \{x \in \mathbb{F}^n \mid \text{perrank}(A \mid x) = m\}$$

is a linear subspace with $\dim V \leq m$.

Proof. If $m = n$, then $V = \mathbb{F}^n$, and the conclusion is immediate. Now we suppose $m < n$. We first note that V is a linear subspace of $\mathbb{F}^n$. Indeed, $\text{perrank}(A \mid x) = m$ means that every $(m+1) \times (m+1)$ submatrix of $(A \mid x)$ has zero permanent. Each such permanent is a linear homogeneous polynomial in the coordinates of x ; the common zero set of such polynomials is closed under addition and scalar multiplication, hence a subspace.

Since $\text{perrank}(A) = m$, there exists an $m \times m$ submatrix of A with nonzero permanent. By permuting rows and columns (which does not affect the property of having nonzero permanent), we may assume that this submatrix occupies the first m rows and the first m columns of A . Let B denote this submatrix, so $\text{Per}(B) \neq 0$.

Consider the projection map

$$\pi : V \longrightarrow \mathbb{F}^m, \quad \pi(x) = (x_1, x_2, \dots, x_m),$$

which sends x to its first m coordinates. We claim that π is injective. Suppose $x, y \in V$ and $\pi(x) = \pi(y)$. Let $z = x - y$. Then $z \in V$ and $z_1 = z_2 = \dots = z_m = 0$. We will show that $z_i = 0$ for every $i > m$. Fix an index $i \in \{m+1, m+2, \dots, n\}$ (such indices exist because $n > m$). Consider the $(m+1) \times (m+1)$ submatrix of $(A \mid z)$ obtained by taking rows $1, 2, \dots, m$ and i , and columns $1, 2, \dots, m$ and $m+1$ (the last column is z). This submatrix has the block form

$$M_i = \begin{pmatrix} & & & z_1 \\ & B & & \vdots \\ & & & z_m \\ a_{i1} & \cdots & a_{im} & z_i \end{pmatrix}.$$

Since $z \in V$, we have $\text{perrank}(A \mid z) = m$, hence $\text{Per}(M_i) = 0$.

Now expand $\text{Per}(M_i)$ along its last column. By multilinearity of the permanent,

$$\text{Per}(M_i) = \sum_{j=1}^m z_j \cdot \text{Per}(B \text{ with the } j\text{-th row replaced by the } i\text{-th row of } A) + z_i \cdot \text{Per}(B).$$

Because $z_1 = \dots = z_m = 0$, the first m terms vanish, leaving

$$0 = \text{Per}(M_i) = z_i \cdot \text{Per}(B).$$

Since $\text{Per}(B) \neq 0$, we conclude $z_i = 0$. This holds for every $i > m$, so $z = 0$, and therefore $x = y$. Hence π is injective.

Consequently,

$$\dim V = \dim \pi(V) \leq \dim \mathbb{F}^m = m.$$

This proves the lemma. □

Corollary 4.5. *Let $m \leq n$ and A be an $n \times m$ random matrix with each entry chosen independently and uniformly from a finite field $\mathbb{F}$. Then*

$$\mathbb{P}(\text{perrank}(A) = m) \geq \mathbb{P}(\text{rank}(A) = m).$$

Proof. In Lemma 4.4, if we replace perrank by rank , then $\dim(V) = m$; so by induction on m we obtain the desired inequality. □

Theorem 4.6. *For any matrix A , $\text{perrank}(A) \geq \text{rank}(A)/2$.*

Proof. Let $r = \text{rank}(A)$. Then A contains a nonsingular $r \times r$ submatrix A' . Since every square submatrix of A' is also a square submatrix of A , we have

$$\text{perrank}(A) \geq \text{perrank}(A').$$

Thus it suffices to prove the result for a nonsingular square matrix. We may therefore assume that A is an $n \times n$ nonsingular matrix.

Let $m = \text{perrank}(A)$. There exists an $n \times m$ submatrix B with full perrank . Let $c_1, \dots, c_{n-m}$ be the remaining columns of A . Since $m = \text{perrank}(A)$, for every i we have

$$\text{perrank}(B \mid c_i) = m,$$

so

$$c_i \in \{x \in \mathbb{F}^n \mid \text{perrank}(B \mid x) = m\}.$$

Moreover, since A is nonsingular, the vectors $c_1, \dots, c_{n-m}$ are linearly independent. Therefore

$$n - m \leq \dim\{x \in \mathbb{F}^n \mid \text{perrank}(B \mid x) = m\} \stackrel{\text{Lemma 4.4}}{\leq} m.$$

Hence $m \geq n/2$, and thus

$$\text{perrank}(A') \geq \text{rank}(A')/2 = r/2.$$

Consequently,

$$\text{perrank}(A) \geq \text{perrank}(A') \geq r/2 = \text{rank}(A)/2.$$

□

Remark. Theorem 4.6 is tight as shown by

$$\begin{pmatrix} I & I \\ I & -I \end{pmatrix}$$

over $\mathbb{F}$ with $\text{char}(\mathbb{F}) \neq 2$.

Theorem 4.7. Let B be an $n \times m$ matrix and let $A \in \text{GL}_n(\mathbb{F})$. If $r = \text{perrank}(B)$, then

$$\text{perrank}(B \mid A) \geq \frac{r + n}{2},$$

where $(B \mid A)$ is the $n \times (m + n)$ matrix obtained by concatenating the columns of B and A .

Proof. If $r = n$, the result is immediate. Otherwise, select r columns of B forming an $n \times r$ matrix C_0 with full permanent rank. Enlarge C_0 with a maximal subset of the columns of A such that the resulting $n \times s$ matrix C has permanent rank s (i.e. $s = \text{perrank}(C)$ and s is maximal).

If $s = n$, we are done, so assume $s < n$. Lemma 4.4 gives

$$W = \{z \in \mathbb{F}^n \mid \text{perrank}([C \ z]) = s\}$$

is a linear subspace with $\dim W \leq s$. By maximality of C , every column of A that was not selected for C must belong to W . There are $s - r$ selected columns of A , and all columns of A span $\mathbb{F}^n$ because A is nonsingular. Hence

$$n \leq (s - r) + \dim W \leq (s - r) + s = 2s - r.$$

Therefore $s \geq (r + n)/2$. Hence $\text{perrank}(B \mid A) \geq s \geq (r + n)/2$. □

Corollary 4.8. For any $n \times n$ nonsingular matrices $A_1, A_2, \dots, A_k$, let

$$(A_1 \mid A_2 \mid \dots \mid A_k)$$

denote the $n \times kn$ matrix obtained by concatenating the columns of the given matrices. Then

$$\text{perrank}(A_1 \mid A_2 \mid \dots \mid A_k) \geq (1 - 2^{-k})n.$$

Proof. Theorem 4.6 gives

$$\text{perrank}(A_1) \geq \frac{n}{2}.$$

Theorem 4.7 gives

$$\text{perrank}(A_1 \mid A_2 \mid \dots \mid A_{t+1}) \geq \frac{\text{perrank}(A_1 \mid A_2 \mid \dots \mid A_t) + n}{2}, \quad 1 \leq t < k.$$

Hence

$$\text{perrank}(A_1 \mid A_2 \mid \dots \mid A_k) \geq (1 - 2^{-k})n.$$

□

Remark. By Corollary 4.8, taking $k = p - 2$, $A_1 = A_2 = \dots = A_{p-2} = A$ and using again Alon's argument, Conjecture 1 is true for $n < 2^{p-2}$.

4.3 Matrices Similar to AJT Matrices

In this section, we call the matrices satisfying Conjecture 1 *AJT matrices*, i.e., there exists a vector $x \in \mathbb{F}^n$ such that both x and Ax are nowhere-zero. Akbari et al. [1] proved that every nonzero matrix is similar to an AJT matrix.

Theorem 4.9. *Every nonzero matrix $A \in M_n(\mathbb{F})$ with $|\mathbb{F}| \geq 3$ is similar to an AJT matrix.*

Proof. Suppose that A is in its rational canonical form, and without loss of generality assume that its $m \times m$ zero block, if it exists, is located in its upper left corner. Any nonzero block of A has the form

$$B = \begin{pmatrix} 0 & 0 & \cdots & 0 & b_1 \\ 1 & 0 & \cdots & 0 & b_2 \\ 0 & 1 & \cdots & \vdots & \vdots \\ \vdots & \vdots & \ddots & 0 & b_{k-1} \\ 0 & 0 & \cdots & 1 & b_k \end{pmatrix}.$$

We consider the following cases.

- (1) The last column of B contains a nonzero element, say b_j . Since B is similar to its transpose $B^\top$, we may replace B by $B^\top$ and choose $x = (1, \dots, 1, c_j, 1, \dots, 1)$, where $c_j \in \mathbb{F}^\times \setminus \{-b_j^{-1} \sum_{i \neq j} b_i\}$.
- (2) The last column of B is zero. Then B is similar to

$$C = \begin{pmatrix} 1 & -1 & \cdots & 0 & 0 \\ 1 & -1 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & \vdots & \vdots \\ \vdots & \vdots & \ddots & 0 & 0 \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}.$$

That is, $C = PBP^{-1}$, where P is the matrix that when applied to B from the left replaces the first row of B with the sum of its first and second rows, and leaves the other rows unaltered. It is easily seen that C is an AJT matrix.

Now, since A is assumed to be block diagonal, we can replace all nonzero blocks on the diagonal of A with their similar AJT versions given in (1) and (2) above, and call the matrix thus obtained $\tilde{A}$. If $m = 0$, $\tilde{A}$ is an AJT matrix. If $m > 0$, consider a nonzero row of $\tilde{A}$, say the i -th row. Let $\tilde{A}_j$ denote the j -th row of $\tilde{A}$. Let Q be the invertible matrix such that $(Q\tilde{A})_j = \tilde{A}_j + \tilde{A}_i$, for every $j, 1 \leq j \leq m$, and $(Q\tilde{A})_k = \tilde{A}_k$, for any $k, m+1 \leq k \leq n$. It is not hard to see that $Q\tilde{A} = Q\tilde{A}Q^{-1}$. We conclude that $Q\tilde{A}Q^{-1}$ is an AJT matrix. $\square$

If Conjecture 1 is true, then every matrix $A \in \text{GL}_n(\mathbb{F}_q)$ with $q > 3$ is an AJT matrix.

4.4 A Rectangular-Matrix Analogue

The nowhere-zero problem can also be considered for rectangular matrices. We first record a simple result that holds over sufficiently large finite fields, which was proved in [1].

Theorem 4.10. *Let A be an $n \times m$ matrix over $\mathbb{F}_q$ with $q > n + 1$. Then A has no zero row if and only if, for every $u \in \mathbb{F}_q^m$ and $v \in \mathbb{F}_q^n$, there exists $x \in \mathbb{F}_q^m$ such that both $x - u$ and $Ax - v$ are nowhere-zero.*

Proof. The necessity is immediate by taking $v = 0$. Conversely, suppose that A has no zero row. There are $(q-1)^m$ vectors $x \in \mathbb{F}_q^m$ such that $x - u$ is nowhere-zero.

Fix a row a_i of A . Since $a_i \neq 0$, choose an index j with $a_{ij} \neq 0$. After the other $m-1$ coordinates of x have been fixed, the equation $a_i x = v_i$ determines at most one value of x_j . Hence at most $(q-1)^{m-1}$ admissible vectors satisfy $(Ax)_i = v_i$.

Thus at most $n(q-1)^{m-1}$ admissible vectors are bad for at least one row. Since $q-1 > n$,

$$(q-1)^m > n(q-1)^{m-1},$$

so at least one admissible vector is good for every row. $\square$

A natural stronger question is whether the size condition on the field can be replaced by the full-row-rank assumption.

Problem 1. Let A be an $n \times m$ ($m > n$) matrix over $\mathbb{F}_q$ with $q > 3$ satisfying $\text{rank}(A) = n$. Must there exist $x \in (\mathbb{F}_q^\times)^m$ such that $Ax \in (\mathbb{F}_q^\times)^n$?

This problem is at least as difficult as the Alon–Jaeger–Tarsi conjecture. Indeed, let $B \in \text{GL}_n(\mathbb{F}_q)$ be arbitrary and consider the full-row-rank matrix

$$A = (B \mid 0) \in \mathbb{F}_q^{n \times (n+1)}.$$

If Problem 1 has an affirmative answer for this matrix, then there exists

$$x = (x_1, \dots, x_n, x_{n+1}) \in (\mathbb{F}_q^\times)^{n+1}$$

such that Ax is nowhere-zero. But

$$Ax = B \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Since $(x_1, \dots, x_n)$ is itself nowhere-zero, this gives a nowhere-zero vector $y \in \mathbb{F}_q^n$ such that By is nowhere-zero. Thus an affirmative solution to Problem 1 would imply the Alon–Jaeger–Tarsi conjecture.

References

- [1] S. Akbari, K. Hassani Monfared, M. Jamaali, E. Khanmohammadi, and D. Kiani. On the existence of nowhere-zero vectors for linear transformations. *Bulletin of the Australian Mathematical Society*, 82(3):480–487, 2010.
- [2] Noga Alon. Combinatorial nullstellensatz. *Combinatorics, Probability and Computing*, 8(1-2):7–29, 1999.
- [3] Noga Alon and Michael Tarsi. A nowhere-zero point in linear mappings. *Combinatorica*, 9(4):393–395, 1989.
- [4] F Jaeger. Problem presented in the 6th hungar. comb. coll., eger, hungary 1981, and: Finite and infinite sets, 1982.
- [5] János Nagy and Péter Pál Pach. The Alon-Jaeger-Tarsi conjecture via group ring identities. *Journal of the European Mathematical Society*, 2025.
- [6] Yang Yu. The permanent rank of a matrix. *J. Comb. Theory A*, 85:237–242, 1999.